

ASSIGNMENT 3:

ENCAPSULATION-PACKAGES, ACCESS MODIFIERS, AND DATA HIDING :

/* Develop a Java-based banking system tailored for the PCCOE Cooperative Bank to facilitate account management and transactions.

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money

*/

```
import java.util.*;
```

```
class Bank {
```

```
    int accno;
```

```
    String name;
```

```
    float bal;
```

```
    Bank(int a,String n,float b){
```

```
        this.accno=a;
```

```
        this.name=n;
```

```
        this.bal=b;
```

```
    }
```

```
    void deposit(Scanner sc) {
```

```
        System.out.println("Enter the amount you want to Deposit: ");
```

```
        float amt = sc.nextFloat();
```

```
        bal = bal + amt;
```

```
        System.out.println("Total Balance = " + bal);
```

```
    }
```

```
    void withdraw(Scanner sc) {
```

```
        System.out.println("Enter the amount you want to Withdraw: ");
```

```
        float amt = sc.nextFloat();
```

```
        if (amt > bal) {
```

```
            System.out.println("Insufficient Balance!");
```

```
        } else {
```

```

        bal = bal - amt;
        System.out.println("Total Balance = " + bal);
    }
}

public class Main {
    public static void main(String args[]) {
        Scanner sc = new Scanner(System.in);
        int ch;
        int i = 1;
        Bank[] b = new Bank[20];
        System.out.print("Enter Account holder name: ");
        String n = sc.nextLine();
        System.out.print("Enter Initial Balance: ");
        float ba = sc.nextFloat();
        sc.nextLine();
        b[0] = new Bank(1,n,ba);
        do {
            System.out.println("\n***** MENU *****");
            System.out.println("1. Add Account");
            System.out.println("2. Remove Account");
            System.out.println("3. Deposit Money");
            System.out.println("4. Withdraw Money");
            System.out.println("5. View Records");
            System.out.println("6. Exit");
            System.out.print("Enter your choice: ");
            ch = sc.nextInt();
            switch (ch) {
                case 1:
                    sc.nextLine();
                    System.out.print("Enter Account holder name: ");
                    String name = sc.nextLine();
                    System.out.print("Enter Initial Balance: ");
                    float bal = sc.nextFloat();
                    b[i] = new Bank(i+1,name,bal);
                    System.out.println("Account Added Successfully.");
                    System.out.println("Account Number = " + b[i].accno);

```

```

        i++;
        break;
case 2:
    System.out.println("Enter Account Number to Remove: ");
    int r = sc.nextInt();
    for (int j = 0; j < i; j++) {
        if (b[j] != null && b[j].accno == r) {
            b[j] = null;
            System.out.println("Account Removed Successfully.");
            break;
        }
    }
    break;
case 3:
    System.out.println("Enter Account Number: ");
    int d = sc.nextInt();
    for (int j = 0; j < i; j++) {
        if (b[j] != null && b[j].accno == d) {
            b[j].deposit(sc);
            break;
        }
    }
    break;
case 4:
    System.out.println("Enter Account Number: ");
    int w = sc.nextInt();
    for (int j = 0; j < i; j++) {
        if (b[j] != null && b[j].accno == w) {
            b[j].withdraw(sc);
            break;
        }
    }
    break;
case 5:
    System.out.println("\n**** Records ****");
    for(int k=0;k<=i;k++){
        if(b[k]!=null){
            System.out.println("\nAccount no."+b[k].accno);

```

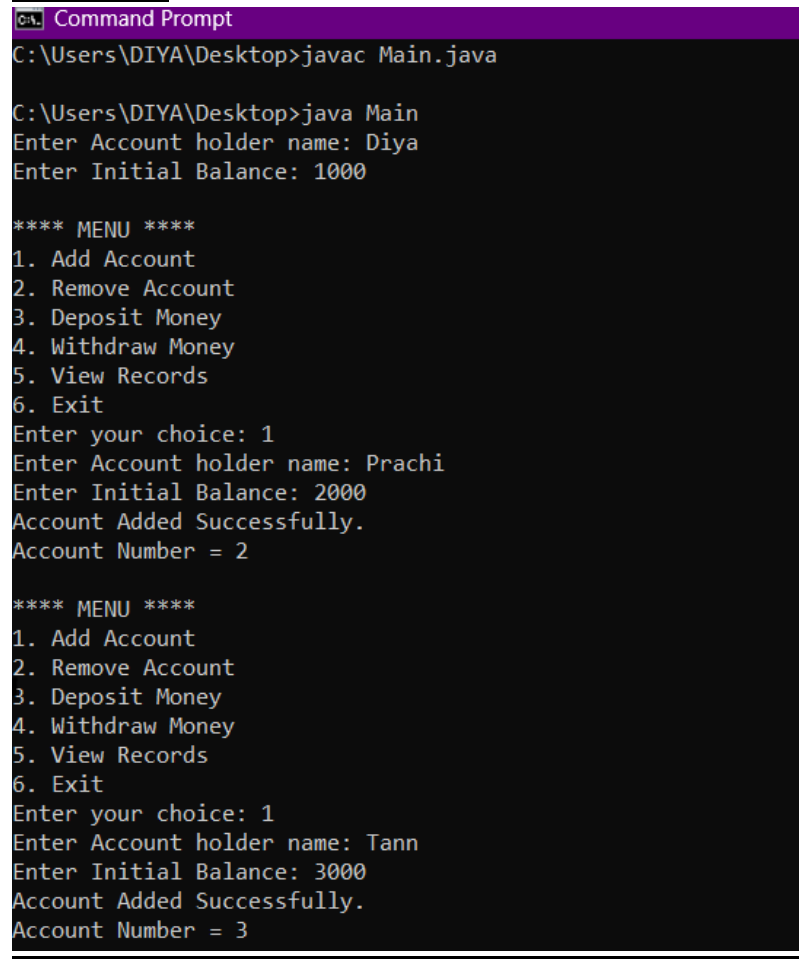
```

        System.out.println("Holder name: "+b[k].name);
        System.out.println("Total Balance: "+b[k].bal);
    }
}
break;
case 6:
    System.out.println("Successfully Exited");
    break;
default:
    System.out.println("Invalid Choice");
}

} while (ch != 6);
}
}

```

OUTPUT:



```

C:\> Command Prompt
C:\Users\DIYA\Desktop>javac Main.java

C:\Users\DIYA\Desktop>java Main
Enter Account holder name: Diya
Enter Initial Balance: 1000

**** MENU ****
1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit
Enter your choice: 1
Enter Account holder name: Prachi
Enter Initial Balance: 2000
Account Added Successfully.
Account Number = 2

**** MENU ****
1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit
Enter your choice: 1
Enter Account holder name: Tann
Enter Initial Balance: 3000
Account Added Successfully.
Account Number = 3

```

Command Prompt

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit

Enter your choice: 2

Enter Account Number to Remove:

2

Account Removed Successfully.

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit

Enter your choice: 5

**** Records ****

Account no.1

Holder name: Diya

Total Balance: 1000.0

Account no.3

Holder name: Tann

Total Balance: 3000.0

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit

Enter your choice: 3

Enter Account Number:

1

Enter the amount you want to Deposit:

2000

Total Balance = 3000.0

C:\ Select Command Prompt

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit

Enter your choice: 4

Enter Account Number:

3

Enter the amount you want to Withdraw:

1000

Total Balance = 2000.0

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit

Enter your choice: 5

**** Records ****

Account no.1

Holder name: Diya

Total Balance: 3000.0

Account no.3

Holder name: Tann

Total Balance: 2000.0

**** MENU ****

1. Add Account
2. Remove Account
3. Deposit Money
4. Withdraw Money
5. View Records
6. Exit

Enter your choice: 6

Successfully Exited

C:\Users\DIYA\Desktop>_